MESTRADO
EM SISTEMAS
DE TRANSPORTES

TS – Transportes e Sustentabilidade

(Transportation and Sustainability)

2026/2027

Project

"Environmental Impacts of the Ubike Long-term Rental Program"

Due 24th October 2026

1. Objectives

This project aims to enhance your capacity to assess the potential to reduce end-of-pipe emissions through mode shifting from all modes of transportation to cycling among the IST community (students, staff, and faculty). This project is based on a case study developed in our lab (http://ushift.tecnico.ulisboa.pt) that included a survey we conducted.

2. Software

The reference software is Excel (or R) and COPERT 5.3 (https://copert.emisia.com/utilities/copert/download/).

3. The case study and data set

The dataset from IST's UBike program is available on the course webpage as a curated, analysis-ready file (cleaned and validated; missing values treated). It comprises 1,270 survey responses from faculty, students, and staff across all IST campuses (Alameda, Tagus Park, CTN Loures). The survey explored the willingness of IST community members to switch to cycling for their commutes if a long-term bike rental system were implemented at IST.

Respondents were asked whether they would use such a system for daily commuting, in addition to providing basic sociodemographic, housing, and mobility patterns/constraints information. The survey included questions on key determinants of daily mobility (e.g., car ownership, monthly transit card, travel time and distance) and cycling-specific conditions (e.g., availability of bike parking at home). By combining sociodemographic characteristics with mobility determinants, the dataset enables estimation of different probabilities of modal shift to cycling and assessment of the credibility of respondents' stated willingness to adopt cycling for commuting.

Questions (variables) included in the survey and presented in the data set.

#	Question	Column in Excel
1	ID	A
2	Who are you? (Student - 1; Faculty or researcher - 2; Staff - 3)	B
3	Which campus of IST? (Campus Tecnológico e Nuclear - 0; Alameda - 1; Tagus Park - 2)	C
4	Age	D
5	Gender (Female - 1)	E
6	Where do you live (Municipality)?	F
7	How many children do you have aged under 1?	G
8	How many children do you have aged between 1 to 5 years?	H
9	How many children do you have aged between 6 to 10 years?	I
10	How many children do you have aged between 11 to 15 years?	J
11	How many children do you have aged more than 15 years old?	K
12	Do you usually come to IST in your private vehicle? (no - 1; yes, with a car - 2; yes, with a motorbike - 3)	L
13	What is your usual travel time to IST from home? (minutes)	M
14	If you don't travel by car to IST, what modes of transportation are combined in your daily commute? (1 - walking only; 2 - Private car; 3 - Carpool; 4 - Bus; 5 - Ferry; 6 - Bike; 7 - Rail; 8 - Metro; 9 - motorbike; 10 - IST Shuttle; 11 - Taxi; 12 - Other)	N
15	Is there any regular intermediate stop in your daily commuting (morning and/or afternoon)? If yes, choose the most common activity (more than twice a week)? (No - 0; Support to children (e.g., drive/walk children to/from school) - 1; Support to older adults - 2; Shopping - 3; Gym/Sports - 4; Work (e.g., working student) - 5; Other - 6)	O
16	Do you have a Transit monthly card? (Yes - 1)	P
17	What is the travel distance of your commuting trip? (km)	Q
18	What is the fuel type of your car (if you use a car to come to IST)? (Diesel or Diesel Hybrid - 1; Petrol or Hybrid petrol - 2; LPG - 3; Electric - 4)	R
19	What type of vehicle do you own or use to come to IST? (City car - 1; Utility - 2; Estate - 3; MPV - 4; SUV - 5; Larger engine (>3 L) - 6; Scooter - 7; Motorbike (>12cc) - 8)	S
20	Are you used to cycling in urban areas (besides segregated bike paths)? (No - 0; Yes - 1; A bit - 2)	T
21	Do you have space to park a bike at home? (No - 0; yes, inside my flat/house - 1; Yes, in the building/garage - 2; Yes, in a sheltered space (balcony, backyard, etc.) - 3; Yes, in an unsheltered space (balcony, backyard, etc.) - 4)	V
22	If you live in a building, which floor do you live on? (note: "0" - ground floor; "-1" - basement; "-2" - sub-basement; "5" - higher than 4th floor)	W
23	Would you consider using a bike-sharing system at IST and shift to cycling for daily commuting? (Not interested - 0; Yes, even if it is not electric - 1; Yes, if it is electric - 2; I have to think - 3)	U

4. Your tasks

Compute the main descriptive statistics of the survey

Adapt and justify the rule-based decision tree provided as a scaffold below (Figure 1) to determine whether each respondent is prone to shift to cycling, irrespective of their stated willingness (question 23). You must (a) explain the rationale of each gate against the Hierarchy of Cycling Needs, (b) set and justify the probability assigned to each leaf, and (c) incorporate at least two of the optional modifier variables (e.g. children, intermediate stops, travel time, campus) as

reasoned ±1-tier adjustments. You are free to refine the thresholds and add branches, provided the logic remains consistent and documented.

Based on a self-determined probability to each of the four possible categories of willingness to shift to cycling (not interested, unsure, interested if electric, interested for all bikes), calculate the km shifted from each mode (walking only; Private car; Carpool; Bus; Ferry; Private Bike; Rail; Metro; Motorbike; IST Shuttle; Taxi) to cycling, for each respondent.

Calculate the emission factors (considering the energy lifecycle) for the substituted fossil-fuelled car and motorbike trips using COPERT and refer to the Excel file for other modes (urban buses, rail, metro, and ferry).

Consider the following gases:

- Main greenhouse gases: CO_2, CH_4, N_2O, and combined CO_{2eq}
- Critical air pollutants: NO_X, PM

Calculate the potential emission savings of daily commuting of each respondent (and summed across community profiles – student, staff, faculty), based on the estimated km shifted from all modes to cycling and the emission factors.

Discuss the results obtained

5. Methodological notes to students

- Please consider using "p-km" as the functional unit for the emission factors. Consequently, they should be expressed as "g/p.km." To calculate emissions per trip, i.e., per passenger, you must consider vehicles' occupancy rate. You can find average occupancy rates in the Excel file.
- If the answer to question 14 indicates a combination of modes, then consider the most impactful mode (based on the estimated emission factors).
- Question 23 lacks a straightforward 'yes' or 'no' answer regarding the stated potential shift from other modes of transportation to cycling. It is therefore necessary to propose a method for assessing the probability of shifting from the current mode to cycling. For instance, an individual who plans to use bicycles daily, lives a considerable distance away from IST, possesses urban cycling experience, resides in a multi-storey apartment building without an lift, and lacks convenient bike storage at home, would have a considerably lower likelihood of switching to cycling compared to someone who also intends to cycle regularly to IST and enjoys ideal conditions (lives up to 5km from IST, has urban cycling proficiency, no family and a house with easy bike storage).

  Once a rule-based decision tree is formulated, probabilities must be assigned to each case, ranging from "high likelihood of shifting " to "low likelihood of shifting." It's important to note that these probabilities are subjective and should be determined by your reasoned judgment, provided they align logically with the decision tree and the adopted criteria. I will respect your judgment if it is logically consistent with the decision tree.

  Ultimately, if a respondent chooses to shift to cycling under favorable conditions, the corresponding potential emissions reduction will be assigned a 100% weighting factor. However, under less-than-ideal conditions, the saved emissions might be assigned a weighting factor of, for instance, 5%. Intermediate probabilities should fit in between.

Example of a Simple Decision Tree with Subjective Probabilities

Gate 1 – Q20: Can the respondent ride? (urban cycling > 0)

- No ⟶ go to Gate 2 (left branch)
- Yes ⟶ go to Gate 3 (right branch)

Gate 2 – Q17: Is distance < 5 km? (for those who *cannot* ride)

- No ⟶ leaf: VERY LOW (P ≈ 0.05)
- Yes ⟶ go to Gate 2a

Gate 2a – Q21 parking & Q22 floor OK? (short trip, cannot ride)

- No ⟶ leaf: LOW (P ≈ 0.20)
- Yes ⟶ leaf: MODERATE (P ≈ 0.40)

Gate 3 – Q17: What is the distance? (for those who *can* ride) – splits three ways:

- < 5 km ⟶ go to Gate 3a
- 5–10 km ⟶ go to Gate 3b
- > 10 km ⟶ go to Gate 3c

Gate 3a – Q20 confident & Q21 parking? (can ride, < 5 km)

- No ⟶ leaf: MODERATE (P ≈ 0.40) *(shown as MOD-HIGH/MODERATE pair)*
- Yes ⟶ leaf: HIGH (P ≈ 1.00) *(with MOD-HIGH ≈ 0.60 as the intermediate outcome)*

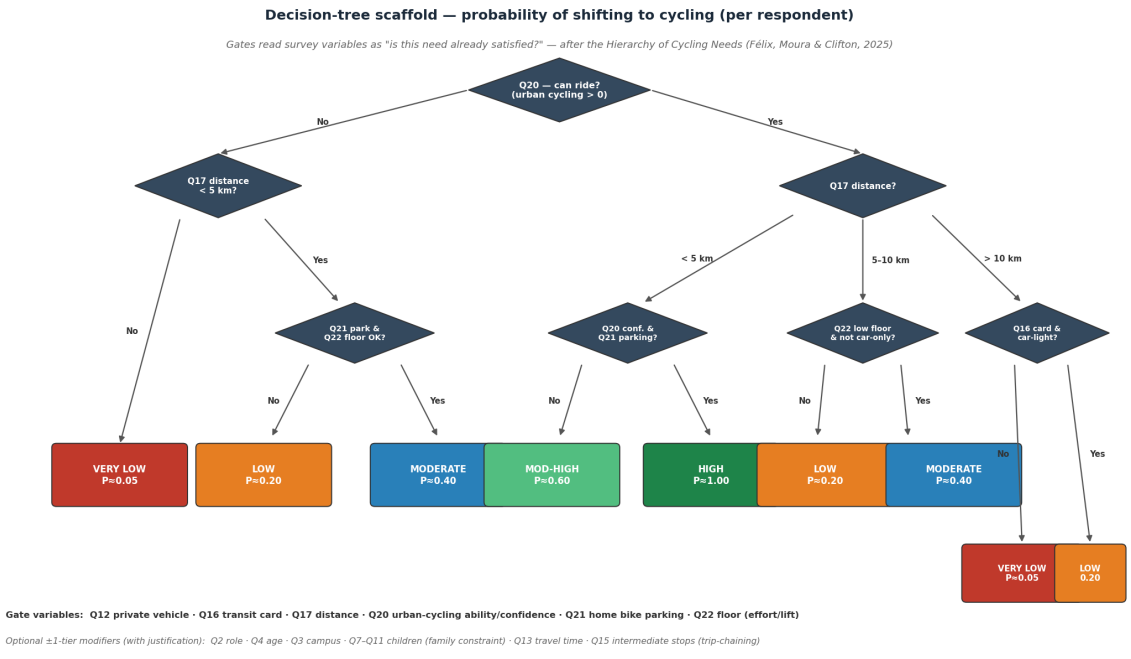
Gate 3b – Q22 low floor & not car-only? (can ride, 5–10 km)

- No ⟶ leaf: LOW (P ≈ 0.20)
- Yes ⟶ leaf: MODERATE (P ≈ 0.40)

Gate 3c – Q16 transit card & car-light? (can ride, > 10 km)

- No ⟶ leaf: VERY LOW (P ≈ 0.05)
- Yes ⟶ leaf: LOW (P ≈ 0.20)

Figure 1 – Decision-tree scaffold to adapt and justify (Task 2)



- How to Compute Final Weighted Emissions

  - Define the Reference Emissions (Current Trip, i.e., before shifting)

- For each respondent, calculate the emissions of their current trip to IST in grams of gas per trip.
- This is done by: $E_{current} = EF_{mode} \times Distance$

where, EF_{mode} - emission factor of the current mode (g/p.km)
$Distance$ - trip distance (km)

– Compute the Maximum Potential Emissions Savings

- From the decision tree, each respondent is assigned a probability of shifting (P_{shift}) to cycling for each likelihood (for example, "high" ≈ 100%, "moderate" ≈ 40%, "low" ≈ 20%, "very low" ≈ 5%).
- Multiply the current emissions ($E_{current}$) by this probability (expressed as a factor between 0 and 1): $E_{saved} = E_{current} \times P_{shift}$

– Interpret the Result

- E_{saved} is the expected emission reduction for each respondent, after considering the likelihood of adopting cycling.
- For example:
  - If a student's current trip emits 1000 g CO_2/trip and they are in a "high likelihood" branch (100%), the savings = 1000 g.
  - If another student's trip emits 1000 g CO_2/trip but they fall under "very low likelihood" (5%), the savings = 50 g.

– Aggregate Across Respondents

- The total weighted emissions savings for IST's community is the sum of the weighted savings across all respondents: $E_{total} = \sum E_{saved}$
- You should provide the aggregate emissions savings by user type (student, staff, faculty). You might also use other disaggregation criteria (e.g., per mode)

6. Reporting

Your grade will be based entirely on the quality of your slide report. The following aspects will be evaluated:

1. Decision Tree Proposal (6/20)
   - Structure of the Tree: The decision tree is logically designed, with meaningful nodes, branches, and outcomes.
   - Justification of Variables: The choice of splits (e.g., by sociodemographic characteristics, housing conditions, mobility constraints) is well motivated by reasoning, literature, or common sense.

What is expected: Students should propose a rule-based decision tree that, in the most reasonable way, describes the conditions under which members of the IST community might shift to cycling, while explaining the rationale behind the decision tree. Each branch should reflect thoughtful consideration of factors such as age, gender, housing type, distance to campus, car ownership, and access to public transport. Keeping the scaffold tree to the bare minimum provided here scores (3/20)

2. Correctness of Calculations and Analysis (7/20)

- Accuracy of statistical or computational work.
- Transparent explanation of any formulas, thresholds, or assumptions.
    - Likelihood: Probabilities assigned to different paths are realistic and coherent.
    - Justification: Students provide reasoning for their probability choices (e.g., why they think a certain profile of student, staff or faculty is more or less likely to adopt cycling).
- Correct handling of the dataset (no inconsistencies or misinterpretations).

What is expected: Students are not required to compute probabilities statistically. Instead, they should use informed judgment to assign probabilities, supported by their understanding of mobility behaviour and the contextual information available (i.e., from the dataset provided).

3. Discussion of Results (5/20)

- Interpretation: Go beyond presenting the decision tree. Explain what the results mean in terms of cycling adoption at IST.
- Determinants: Identify which factors are most influential in predicting the modal shift.
- Credibility Check: Discuss whether respondents' stated willingness aligns with realistic mobility and sociodemographic constraints.
- Environmental implications: Discuss the before and after variation of air pollutant emissions, resulting from the potential shift to cycling.
- Implications: Reflect on what the results suggest for the potential success of a bike-sharing system at IST, including barriers and opportunities.
- Limitations: Acknowledge any shortcomings in the dataset, assumptions, or methods.

What is expected: Students should not only describe their findings but also critically assess them, connecting results to mobility theory, behavioral aspects, and practical implementation issues.

4. Report Quality (2/20)

- Slides must be well-structured, readable, and consistent in style.
- Use of tables, graphs, or visual representations where appropriate.
- References properly cited (Harvard style) and listed at the end.